ALEJANDRO VELOZ BARCENAS

Senior Software Engineer | .NET | Modernization | AI-Assisted Engineering

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PROFESSIONAL SUMMARY

Senior Software Engineer with 12+ years of experience designing and modernizing enterprise software across fintech, supply chain, financial services, and other business-critical environments. Strong background in C#/.NET, SQL Server, REST APIs, legacy modernization, data migration, ETL workflows, and full-stack development.

Developed a Claude Code methodology to coordinate specialized agents, preserve project context, and incorporate tests and QA/security reviews. Applied it to Marvin, Easy Quotes, and 3Dimensional while retaining responsibility for product, architecture, and delivery.

CORE TECHNOLOGIES

Backend: C#, .NET Core, .NET Framework, ASP.NET, REST APIs, Go, Ruby on Rails, Entity Framework, Dapper

Data & ETL: SQL Server, T-SQL, Azure Data Factory, Azure Blob Storage, PostgreSQL, Redis, data migration and validation

Frontend: Angular, TypeScript, JavaScript, React, Next.js, HTML, CSS

Architecture: Clean Architecture, SOLID, modular monoliths, asynchronous processing, legacy modernization

DevOps & Quality: Docker, Azure DevOps, GitHub Actions, Jenkins, SonarQube, Datadog, unit and integration testing

AI: Claude Code, OpenAI Codex, GitHub Copilot, OpenAI API; own methodology for agent-assisted development

PROFESSIONAL EXPERIENCE

ENDAVA

Senior Software Engineer | February 2025 - September 2026

Worked across enterprise data migration and modernization initiatives involving SAP S/4HANA, Azure Data Factory, SQL Server, legacy applications, and .NET Core.

SAP Master Data Migration

- Executed, monitored, and troubleshot Azure Data Factory ETL pipelines supporting enterprise master data migration into SAP S/4HANA.
- Investigated migration failures using SQL queries, ADF pipeline runs, validation results, and migration logs.
- Developed SQL transformations, data corrections, mapping rules, cross-reference logic, and validation processes to resolve migration and data-quality issues.
- Worked with Azure Data Factory, SQL Server, Azure Blob Storage, SharePoint, and enterprise migration workflows.
- Supported data domains including Material Master, Vendors, Purchasing Info Records, and HTS Commodity Codes.

Legacy Application Modernization

- Analyzed legacy Visual Basic processes, business rules, and SQL Server interactions and migrated functionality into modern .NET Core REST APIs.
- Applied Clean Architecture principles and implemented repositories, DTOs, application components, database integrations, and automated tests.
- Used Docker and Swagger/OpenAPI to support development, testing, documentation, and deployment workflows.

Warehouse Backend Modernization

- Led analysis and refactoring of complex SQL Server stored procedures supporting high-volume warehouse automation operations.
- Redesigned data mutation and validation workflows to improve data integrity, predictability, and maintainability.
- Contributed to .NET Core architectural improvements, database-layer redesign, and SQL Server-based asynchronous processing design for future scalability.

CONEKTA

Senior Backend Developer | May 2022 - September 2024

- Led a software project from planning and technical design through implementation and delivery for business-critical checkout and payment processing.
- Modernized the architecture of a legacy C# API using updated design patterns to reduce technical debt and improve maintainability.
- Redesigned and maintained Golang backend services and contributed to data-model improvements supporting checkout operations.
- Built unit tests for critical payment flows and worked with Docker, Jenkins, GitHub Actions, SonarQube, and Datadog.

ENDAVA

Senior Software Developer | February 2021 - May 2022

- Developed unit and integration testing strategies for software controlling automated warehouse systems.
- Created a simulator to reproduce automation workflows and physical-system interactions in controlled environments.
- Helped establish a GitFlow-based development and testing process; worked with C#, WPF, WCF, SQL Server, and onsite production support.

ACCENTURE

Senior Application Development Specialist | August 2019 - February 2021

- Participated in the modernization of a legacy Silverlight mortgage application toward a modern .NET Core architecture.
- Developed new modules using Angular, TypeScript, C#, and SQL Server while maintaining legacy functionality during migration.
- Worked directly with client and management stakeholders on technical priorities and quarterly delivery planning.

BSD ENTERPRISE

Web Developer | June 2018 - July 2019

- Owned two major functional areas of a new insurance platform from implementation through production support.
- Developed .NET backend services and JavaScript/jQuery frontend functionality; designed APIs and maintained WCF integrations.
- Coordinated a small development team plus testing and deployments across environments.

DEFINITY FIRST

Web Systems Engineer | 2014 - 2018

- Developed and maintained .NET and JavaScript systems supporting time-critical emergency service workflows.
- Helped migrate real-time communications from polling to SignalR and integrated Google Maps JavaScript APIs.
- Developed SQL Server processes and worked with SendGrid and Mailgun integrations.

SELECTED PRODUCTS & PROJECTS

Marvin | usamarvin.com

Multi-tenant SaaS in production for service businesses, covering appointments, customers, payments, and reporting. Designed a modular monolith and owned product, architecture, full-stack development, data, infrastructure, and deployment.

Technologies: Next.js, Node.js, PostgreSQL, Redis, Docker

Easy Quotes | easy-quotes.com

Multi-tenant SaaS in production for quoting, approvals, and payment tracking. Designed a modular monolith and owned product, architecture, implementation, and deployment.

Technologies: TypeScript, React, Fastify, PostgreSQL, Drizzle, Docker, GitHub Actions

3Dimensional | 3dimensional.mx

Created the platform for his own 3D printing business: visual customization, PDF quotes and quote history, and internal tools generating 3MF manufacturing files. Owned product and technical decisions, with Claude Code-assisted development.

Technologies: JavaScript, Canvas, Supabase, geometry and 3MF file generation

EDUCATION & LANGUAGES

Universidad de Monterrey - Computer Science Engineering studies

Languages: Spanish (Native) | English (C1 / Advanced)